

ResearchPilot

Autonomous Web Research & Report-Generation Agent

The Problem

- ✗ Non-trivial research needs multiple searches.
- ✗ One-shot LLM responses can stop too early.
- ✗ Answers may rely on incomplete evidence.

The Solution

- ✓ Autonomously plans and evaluates evidence.
- ✓ Maintains state across the workflow.
- ✓ Uses real tools, not just single prompts.

Plan



Act



Observe



Critique



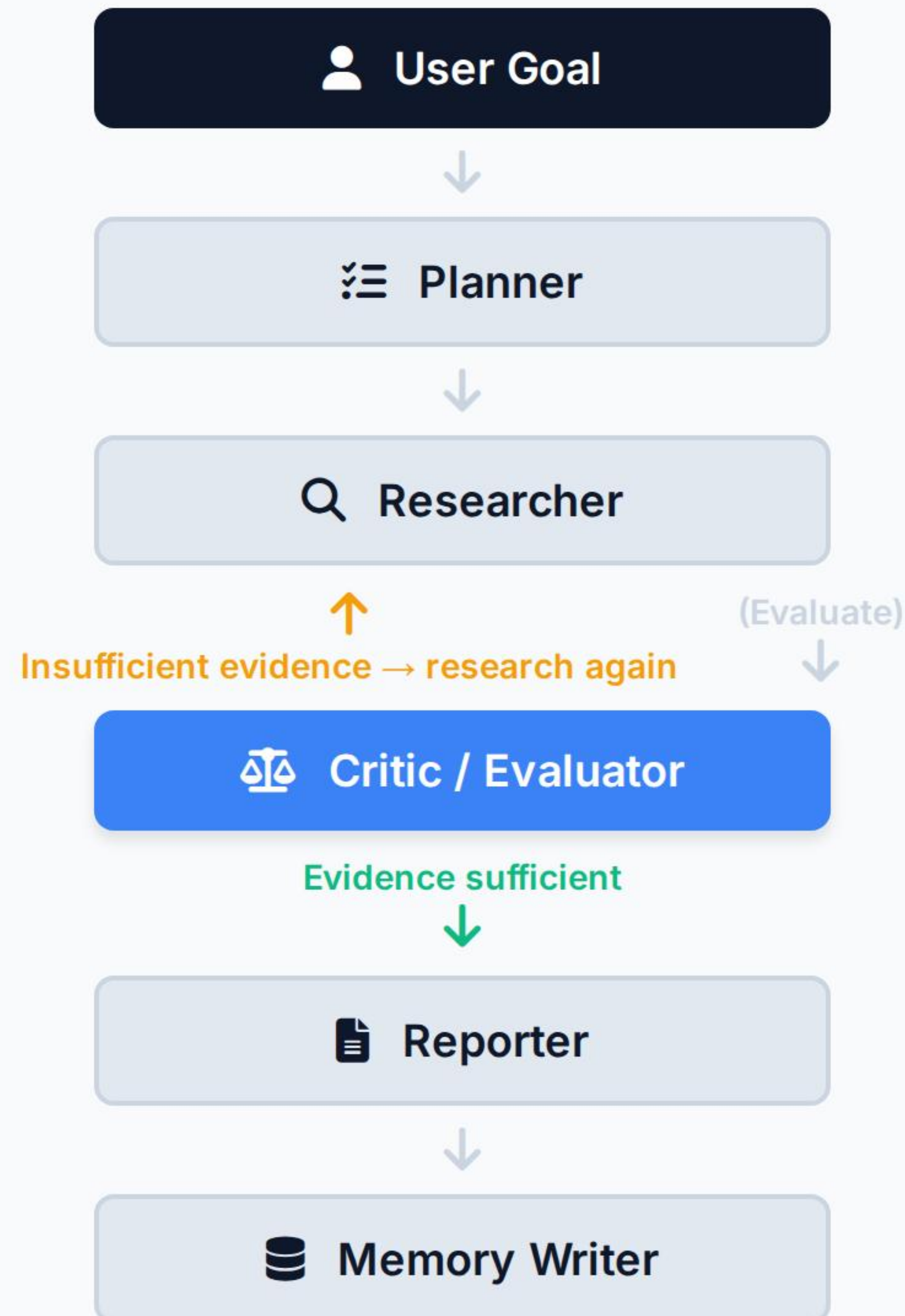
Re-plan



Report

System Architecture

LangGraph-based orchestration with shared AgentState and configurable LLM providers



Agentic Workflow & Tools

7-STAGE WORKFLOW

- PLAN** ● LLM creates 3–5 logical research steps.
- ACT** ● Executes required tool for current step.
- OBSERVE** ● Captures search, page, or math results.
- STATE UPDATE** ● Stores findings, sources, and history.
- CRITIQUE** ● Checks coverage and evidence quality.
- RE-PLAN** ● Adds research when evidence is thin.
- RESPOND** ● Generates structured Markdown report.

IMPLEMENTED TOOLS



web_search

General web research via DuckDuckGo.



page_reader

Reads deep text from HTML and PDFs.



calculator

Safe AST-based arithmetic evaluation.



report_writer

Saves timestamped Markdown reports.

Demo, Memory & Evaluation

> "What is the current pricing for Anthropic's Claude API, including the newest models?"

[PLAN] → [TOOL] → [OBSERVE] → [STATE] → [CRITIC]

IF INSUFFICIENT:

← [RE-PLAN] → [TOOL] → [OBSERVE] →

[CRITIC]

IF SUFFICIENT:

← [REPORT] → [MEMORY]

Lightweight Memory

- Stores successful and failed web-search queries.
- Stores useful source domains.
- Reuses related past experience as planning hints.

Uses previous-run memory to improve future planning.

Deterministic Evaluation

Relevance
1.0000

Completeness
0.9583

Source Cov.
1.0000

Grounding
0.9167

Format Qual.
1.0000

Overall
0.9750

** Fixture-suite diagnostics, not general model accuracy.*

Contest Requirements Checklist

- ✓ Accepts a text-based task
- ✓ Breaks the task into logical steps
- ✓ Uses LLM reasoning for planning
- ✓ Executes real tools and external data access
- ✓ Maintains state and context across steps
- ✓ Supports conditional workflow and re-planning
- ✓ Returns a coherent final output
- ✓ Uses previous-run memory to improve planning

ResearchPilot demonstrates genuine agentic behaviour — not just a single LLM prompt-response.